



Chapter 1 — Substrate

What a language model is, and what one run of it computes

The answer arrives a piece at a time

animation

a question typed, and the reply appearing left to right at natural speed

1600 x 422 px

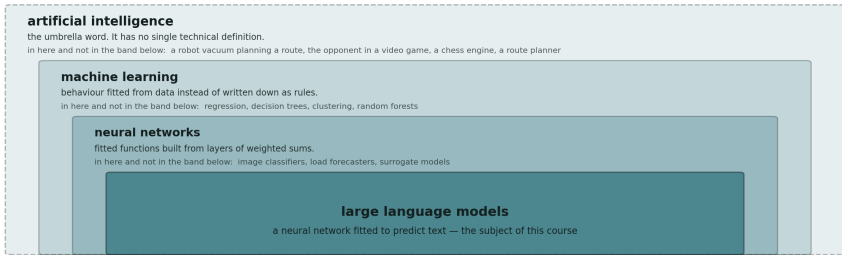
The text you send is the **prompt**. It writes left to right, in pieces — and if you ask the same question twice, the two answers differ.

[--] *Direct observation, made by running it. No external source.*

Artificial intelligence names a family of methods, and language models are one branch of it

Every band contains the one inside it. Not all artificial intelligence is a language model.

dashed = a boundary drawn by usage, not by definition — "artificial intelligence" is applied inconsistently

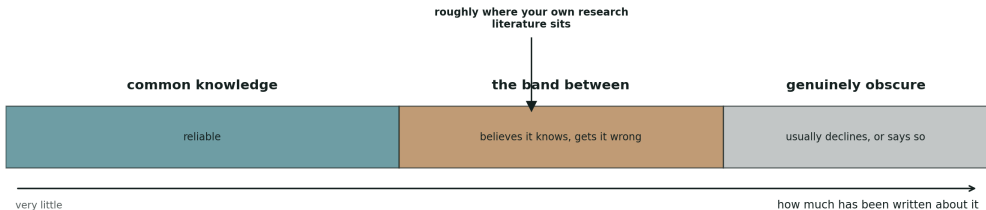


Large language model — a neural network fitted to predict text. Shortened to **LLM**.

[--] *Definition — this term is introduced by the course. No external source.*

It is reliable where a great deal has been written, and unreliable where little has

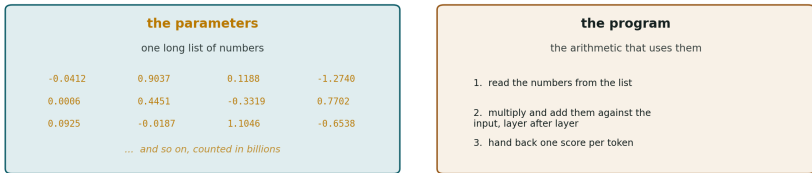
Reliability tracks how much has been written, not how important the topic is.



The corpus decides where it is strong. Nobody has trained a model mainly on your literature.

[--] Definition — this term is introduced by the course. No external source.

A model is a long list of numbers together with the program that reads them



Both are files on a disk. Together they are the model.

Parameters

also called **weights**

Network

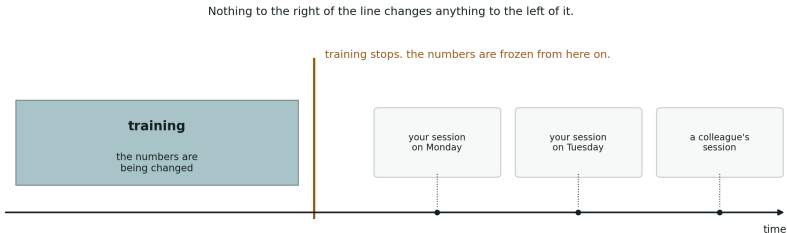
the layered arithmetic

Size

how many parameters

[--] Definition — this term is introduced by the course. No external source.

Training set those numbers and then stopped



Training

the numbers are being changed

Inference

running the finished numbers on
your text

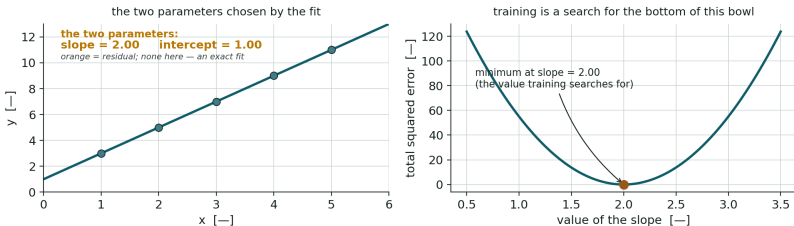
Knowledge cutoff

the date training stopped

[--] Definition — this term is introduced by the course. No external source.

Training searches for the numbers that make the total error smallest

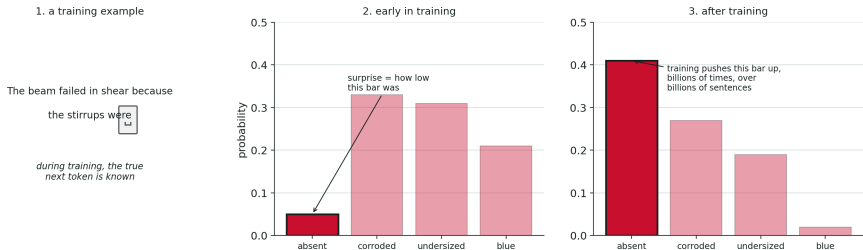
$y = 2x + 1$ exactly — chosen so slope 2.00 and intercept 1.00 can be verified by hand. A real fit does not land on zero residual.



Two parameters here: **slope 2.00** and **intercept 1.00**. A language model does the same search over billions of them, and the data is text.

[--] Computed in this repository — see `assets/figures/gen-figures.py`. No external source.

What training minimises is how surprised the model was by the token that actually came next

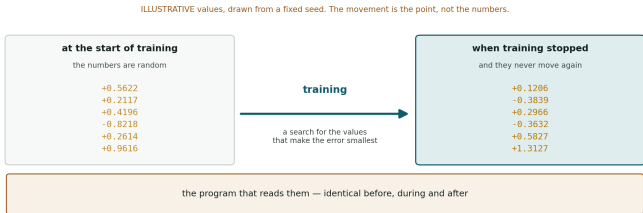


You cannot define "surprised" without a probability. The distribution is what training shaped.

You cannot define *surprised* without a **probability**. So the thing training shaped is a distribution over every token in the list.

[--] Follows from earlier slides in this chapter. No external source.

Training moved these numbers, and nothing else about the system changed



Neural network

layers of multiply-and-add,
named after a biological analogy

Learning

the search you just watched

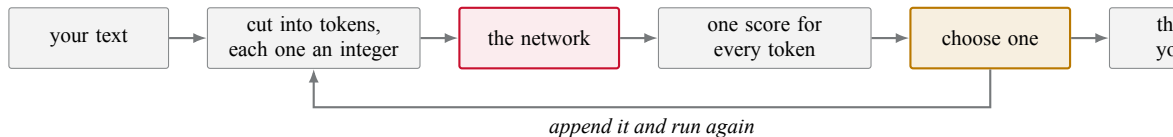
The brain comparison

correlational; the appendix
carries what it establishes

[V] Schrimpf, M., Blank, I. A., Tuckute, G., et al., 2021. *PNAS* 118(45):e2105646118, DOI 10.1073/pnas.2105646118.

[V] Hadidi, N., Fegghi, E., Song, B. H., Blank, I. A. & Kao, J. C., 2026. *Nature Communications* 17(1):5769, DOI 10.1038/s41467-026-72253-7.

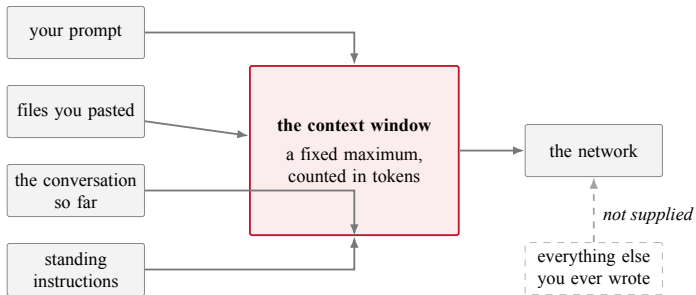
The whole path runs from text to numbers to scores to one piece of text, and repeats



Every box gets its own slide, in this order. **Token** — a chunk of characters from a fixed list, defined next.

[--] Definition — this term is introduced by the course. No external source.

Everything the model is given sits in the context window

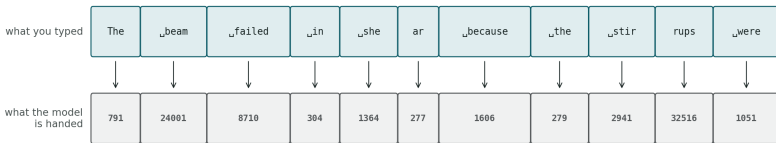


Re-read in full on every pass. **Standing instructions** — text placed in the window at the start of every session.

[--] Definition — this term is introduced by the course. No external source.

A token is a chunk of characters drawn from a fixed list settled before training

SCHEMATIC — no tokeniser was run. Capture a real split before delivery.



marks a leading space, which belongs to the token. "shear" and "stirrups" split mid-word — eleven token IDs, in slate, for nine words.

* Not only text. An image or a PDF page is cut into small squares and turned into numbers in the same window.

The vocabulary

tens of thousands of entries

Token IDs

labels, not quantities

No letters

the network is handed integers

[--] Schematic drawn for this course. Values illustrative and labelled on the figure.

Substrate

Formation

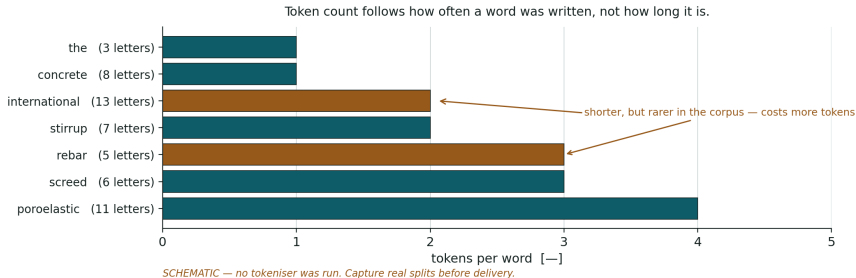
Control

Measure

Failure

Limits

Token count follows how often a word was written, not how long it is



Billed
per token

Context limit
counted in tokens

Speed
quoted per token

[--] Schematic drawn for this course. Values illustrative and labelled on the figure.

Substrate

Formation

Control

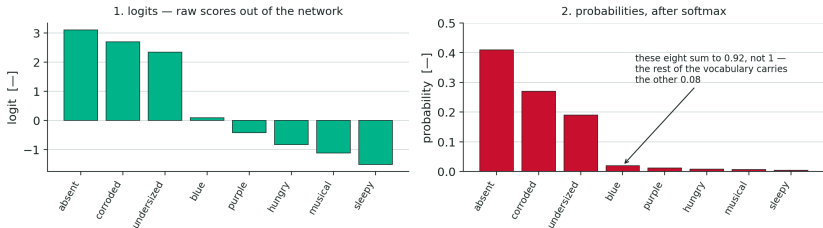
Measure

Failure

Limits

The network returns one score for every token in the list

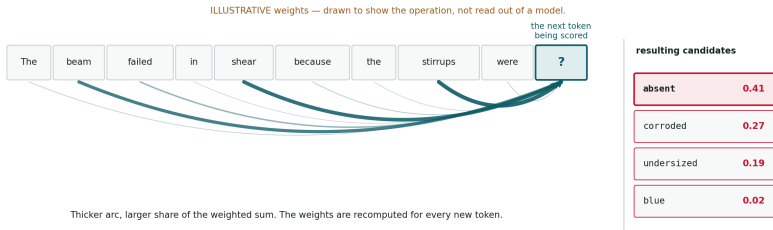
illustrative logits — the softmax transform applied to them is exact, over these eight plus one term for the rest of the vocabulary



The raw scores are **logits**. One run of the network over the window is one **pass**. The four shown do not add to one — the rest of the vocabulary carries the remainder.

[--] Definition — this term is introduced by the course. No external source.

Attention decides which earlier tokens the next score leans on



animation A3 replaces this figure — 1600 × 600 px

A **weighted sum**: the weight on each earlier position comes from comparing it against the position being scored. **Transformer** — the arrangement of arithmetic built from it.

[V] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L. & Polosukhin, I., 2017. *Attention Is All You Need*, arXiv:1706.03762.

Softmax turns the scores into probabilities that add up to one

$$p_i = \frac{\exp\left(\frac{\text{score for token } i}{\text{temperature}}\right)}{\underbrace{\sum_j \exp(z_j/T)}_{\text{sum over every token, so the result adds to 1}}}$$

Worked case — restricted to two candidates so the sum can be checked by hand.

$$z_{\text{absent}} = 2.0 \quad z_{\text{corroded}} = 1.0$$

$$p = \frac{e^{2.0}}{e^{2.0} + e^{1.0}} = \frac{7.389}{10.107} = 0.731$$

All quantities dimensionless.

[--] Computed in this repository — see `assets/figures/gen-figures.py`. No external source.

Taking the top-scoring token every time makes the text repeat

animation

temperature zero, run long enough that the output falls into a loop

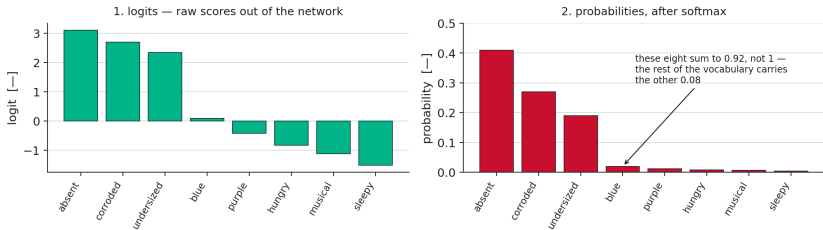
1600 x 422 px

Same input, same scores, same top token — so a recurring phrase recreates the context that produced it. Taking the top every time is **greedy** selection.

[--] *Follows from earlier slides in this chapter. No external source.*

So a token is drawn from the distribution rather than taken from the top

illustrative logits — the softmax transform applied to them is exact, over these eight plus one term for the rest of the vocabulary



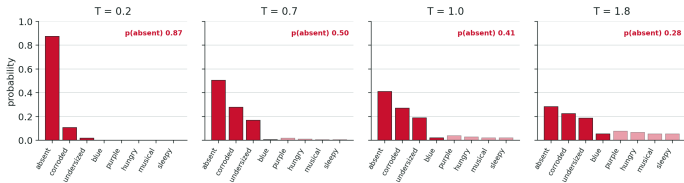
animation A4 replaces this figure — 1600×600 px — the bars must be identical on every draw

The bars are computed from the frozen numbers and never change. The **draw** does.

[--] Follows from earlier slides in this chapter. No external source.

Temperature sets how sharply the top score wins

Same eight candidates, four temperatures — logits are log(table probability), so softmax at $T = 1$ reproduces the table exactly



animation A5 replaces this figure — 1600 × 450 px

Low T

one token takes almost
everything

T = 1

the scores used as they are

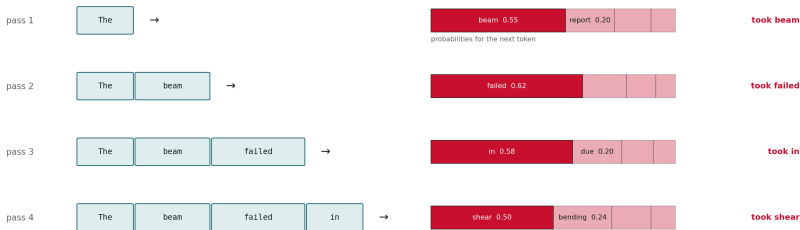
High T

the **probability** spreads

[--] Computed in this repository — see `assets/figures/gen-figures.py`. No external source.

Each chosen token is appended and the whole program runs again

ILLUSTRATIVE — the same frozen numbers are read on every pass. Only the text handed to them has grown.

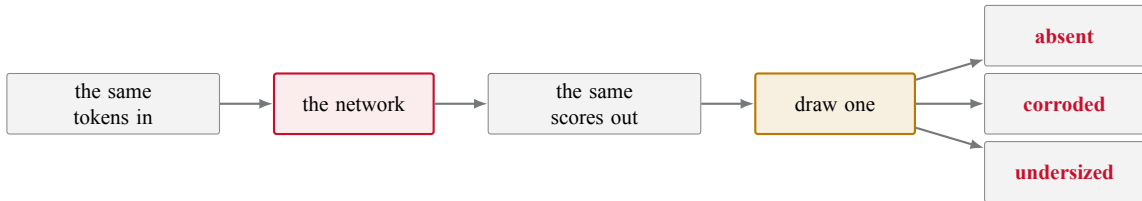


animation A7 replaces this figure — 1600×450 px

Four passes here. A paragraph is a few hundred. The **parameters** are unchanged on every one of them.

[--] Schematic drawn for this course. Values illustrative and labelled on the figure.

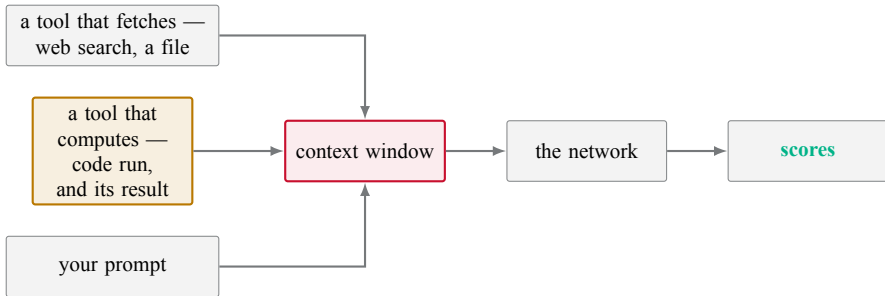
Two runs of the same prompt give the same scores and different answers



The puzzle from the first slide, resolved. Everything left of **draw one** is fixed.

[--] *Follows from earlier slides in this chapter. No external source.*

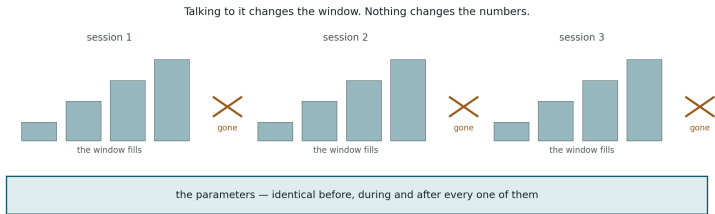
Tools write into the window and can run code beside the network



A **tool** is a program the system runs alongside the network. Its result is written into the window as tokens, and the pass that follows is the one already described.

[V] Anthropic, 2026. *Code execution tool*, platform.claude.com — vendor documentation, fetched 2026-08-20.

The window is discarded when the session ends and the numbers are left as they were



Discarded

the window, the conversation, anything you
pasted

Unchanged

the **parameters**, in every session

[--] Definition — this term is introduced by the course. No external source.

What we covered, and what is still open

What we covered

- ▶ A model is a list of **numbers** and a program that reads them
- ▶ Training fixed those numbers and stopped
- ▶ What it minimised was surprise, so the output is a **distribution**
- ▶ Text is cut into **tokens** from a fixed list
- ▶ Every pass scores every token, then draws one
- ▶ It sees only the context window, and discards it

What is still open

- ▶ Nothing in scoring text makes a thing that answers a question, follows an instruction, or declines a request. What was added?
- ▶ Why does it agree with you so readily?
- ▶ If it is reliable where much has been written, how do you tell which regime you are in?
- ▶ How would you ever know a change to your prompt actually helped?

[--] Follows from earlier slides in this chapter. No external source.